

Java Backend Interview Questions

Mastercard • Deutsche Bank • HSBC • Citi • PhonePe • Razorpay

Target: 27-30 LPA | Java 17 + Spring Boot 3.3.5 | 5 YOY

Answer Formula: WHAT + HOW + WHY

Every answer needs all three. WHAT shows knowledge. HOW shows depth. WHY shows senior thinking.

13

Topic Areas

170+

Questions

3

Priority Levels

30 LPA

Target

Study Priority Guide

■ MUST KNOW PERFECTLY

Spring Security + JWT	<i>You built it — explain full flow</i>
Redis Caching	<i>Shared cache layer across instances</i>
Optimistic Locking	<i>WHERE version=? SQL + 409 response</i>
REST API Design	<i>Status codes, idempotency, pagination</i>
JPA + Hibernate	<i>N+1, lazy loading, transactions</i>
Behavioural Questions	<i>STAR format + your work experience</i>

■ MUST KNOW WELL

Core Java	<i>Streams, generics, concurrency, SOLID</i>
Spring Batch	<i>Chunk processing, job history, restart</i>
Kafka	<i>Topics, partitions, consumer groups</i>
System Design	<i>Payment system, rate limiter, CAP theorem</i>

■ GOOD TO KNOW

Microservices	<i>Gateway, Eureka, circuit breaker</i>
Kubernetes	<i>Pods, deployments, HPA, rolling updates</i>
AWS	<i>EKS, RDS, ElastiCache, MSK, IAM</i>

■ CORE JAVA

- 01 What is the difference between HashMap and ConcurrentHashMap?
- 02 What is the volatile keyword?
- 03 What is the difference between synchronized and ReentrantLock?
- 04 What are functional interfaces?
- 05 What is the difference between Comparable and Comparator?
- 06 What is String immutability and why does it matter?
- 07 What is the difference between checked and unchecked exceptions?
- 08 What is Java memory model? Heap vs Stack?
- 09 What is garbage collection? Which GC algorithms exist?
- 10 What are generics and why use them?
- 11 What is the difference between == and equals()?
- 12 What is Optional and why use it?
- 13 What are streams? Give examples of map, filter, reduce
- 14 What is the difference between parallelStream and stream?
- 15 What is method reference?
- 16 What is a record in Java 17?
- 17 What is a sealed class?
- 18 What is the difference between abstract class and interface?
- 19 Explain the SOLID principles with examples
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■ SPRING BOOT

- | | |
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| 01 | What is Spring Boot auto-configuration? |
| 02 | What is the difference between <code>@Component</code> , <code>@Service</code> , <code>@Repository</code> ? |
| 03 | What is <code>@SpringBootApplication</code> ? |
| 04 | What is dependency injection? Constructor vs setter vs field? |
| 05 | Why constructor injection over field injection? |
| 06 | What is the Spring Bean lifecycle? |
| 07 | What is <code>@Transactional</code> and how does it work? |
| 08 | What is transaction propagation? <code>REQUIRED</code> vs <code>REQUIRES_NEW</code> vs <code>NESTED</code> ? |
| 09 | What is the difference between <code>@SpringBootTest</code> and <code>@WebMvcTest</code> ? |
| 10 | What is <code>@DataJpaTest</code> ? |
| 11 | What is AOP? Where did you use it? |
| 12 | What is the difference between <code>@MockBean</code> and <code>@Mock</code> ? |
| 13 | What is <code>ApplicationContext</code> ? |
| 14 | What is <code>BeanFactory</code> vs <code>ApplicationContext</code> ? |
| 15 | How does Spring Boot handle exceptions? |
| 16 | What is <code>@ControllerAdvice</code> ? |
| 17 | What is the difference between <code>@RestController</code> and <code>@Controller</code> ? |

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16	What is @ControllerAdvice?
17	What is the difference between @RestController and @Controller?

■ SPRING SECURITY + JWT

- 01 What is authentication vs authorization?
- 02 How does JWT work? Explain header, payload, signature
- 03 Why JWT over sessions for microservices?
- 04 What is stateless authentication?
- 05 What is CSRF and why did you disable it for REST APIs?
- 06 What is BCrypt and why hash passwords?
- 07 What is UserDetails and UserDetailsService?
- 08 What is SecurityFilterChain?
- 09 What is OncePerRequestFilter?
- 10 What is the difference between 401 and 403?
- 11 What is @PreAuthorize?
- 12 What is @EnableMethodSecurity?
- 13 How do you invalidate a JWT token before expiry?
- 14 What is the difference between HS256 and RS256?
- 15 Where should JWT secret be stored in production?
- 16 What is OAuth 2.0? How is it different from JWT?
- 17 How do you prevent SQL injection in Spring?
- 18 What is CORS and how do you handle it?
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■ ■ JPA + HIBERNATE

- 01 What is the difference between Entity and domain object?
- 02 What is lazy vs eager loading?
- 03 What is the N+1 problem and how do you fix it?
- 04 What is @Transactional isolation levels?
- 05 What is optimistic vs pessimistic locking?
- 06 What is @Version and how does it work?
- 07 What is the difference between save() and saveAndFlush()?
- 08 What is dirty checking in Hibernate?
- 09 What is first level cache vs second level cache?
- 10 What is JPQL vs native query?
- 11 What is the difference between @OneToMany and @ManyToOne?
- 12 What is CascadeType?
- 13 What is FetchType?
- 14 What is Flyway and why use it?
- 15 Why should you never edit existing migrations?
- 16 What is ddl-auto=validate?
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■ REST API DESIGN

- 01 What are REST principles?
- 02 What is idempotency? Which HTTP methods are idempotent?
- 03 What is the difference between PUT and PATCH?
- 04 What HTTP status codes for: created, conflict, not found, unauthorized, forbidden?
- 05 What is HATEOAS?
- 06 How do you version an API?
- 07 What is pagination and how do you implement it?
- 08 What is the difference between @RequestParam and @PathVariable?
- 09 How do you handle validation errors?
- 10 What is Swagger/OpenAPI?
- 11 What is the difference between POST and PUT?
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■ REDIS + CACHING

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| 01 | What is Redis? |
| 02 | What is the difference between Redis and in-memory cache? |
| 03 | What is cache eviction? What is TTL? |
| 04 | What is @Cacheable and @CacheEvict? |
| 05 | What is cache aside pattern? |
| 06 | What is write through vs write behind caching? |
| 07 | What is cache stampede? How do you prevent it? |
| 08 | What Redis data structures do you know? |
| 09 | What is cache inconsistency in microservices? How do you solve it? |
| 10 | What is distributed locking with Redis? |
| 11 | When would you NOT cache something? |
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■ MULTITHREADING + ASYNC

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| 01 | What is the difference between Thread and Runnable? |
| 02 | What is ExecutorService? |
| 03 | What is ThreadPoolTaskExecutor? |
| 04 | Why not SimpleAsyncTaskExecutor in production? |
| 05 | What is @Async and how does it work? |
| 06 | What is CompletableFuture? |
| 07 | What is the difference between thenApply and thenCompose? |
| 08 | What is a race condition? |
| 09 | What is deadlock? How do you prevent it? |
| 10 | What is volatile? |
| 11 | What is ThreadLocal? |
| 12 | What happens when @Async throws an exception? |
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■ SPRING BATCH

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| 01 | What is Spring Batch? |
| 02 | What is Job, Step, ItemReader, ItemProcessor, ItemWriter? |
| 03 | What is chunk processing? Why not load everything at once? |
| 04 | What is JobRepository? |
| 05 | What is JobLauncher and RunIdIncrementer? |
| 06 | How do you restart a failed job? |
| 07 | What is the difference between @Scheduled and Spring Batch? |
| 08 | When would you use Spring Batch over Kafka? |
| 09 | What is partitioned processing? |
| 10 | What is skip and retry in Spring Batch? |
| 11 | What is the BATCH_JOB_EXECUTION table? |
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- 02 What is a topic, partition, offset?
- 03 What is a producer and consumer?
- 04 What is a consumer group?
- 05 What is at least once vs at most once vs exactly once delivery?
- 06 What is a dead letter queue?
- 07 What is an idempotent consumer?
- 08 What is Kafka retention?
- 09 Why Kafka over RabbitMQ?
- 10 What is the difference between Kafka and @Async?
- 11 What happens when a consumer crashes?
- 12 What is rebalancing?
- 13 What is a Kafka key and why does it matter?
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■ MICROSERVICES

- 01 What is a microservice?
- 02 What is the difference between monolith and microservices?
- 03 What is API Gateway?
- 04 What is service discovery? What is Eureka?
- 05 What is circuit breaker? What is Resilience4j?
- 06 What is the difference between sync and async communication?
- 07 What is OpenFeign?
- 08 What is the Saga pattern?
- 09 What is the difference between Saga choreography and orchestration?
- 10 What is database per service?
- 11 What is eventual consistency?
- 12 What is the strangler fig pattern?
- 13 How do you handle distributed transactions?
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SYSTEM DESIGN

- 01 Design a payment system
- 02 Design a notification system
- 03 Design a rate limiter
- 04 Design a URL shortener
- 05 How would you handle 10 million transactions per day?
- 06 How would you prevent double spend?
- 07 What is CAP theorem?
- 08 What is eventual consistency?
- 09 What is sharding and consistent hashing?
- 10 What is the difference between SQL and NoSQL?
- 11 What is CQRS?
- 12 What is event sourcing?
- 13 How do you design for high availability?
- 14 What is the difference between L4 and L7 load balancing?
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■ ■ KUBERNETES + AWS

- 01 What is a Pod?
- 02 What is a Deployment vs StatefulSet?
- 03 What is a Service in Kubernetes? ClusterIP vs NodePort vs LoadBalancer?
- 04 What is an Ingress?
- 05 What is HPA (Horizontal Pod Autoscaler)?
- 06 What is a ConfigMap vs Secret?
- 07 What is a namespace?
- 08 What is a liveness vs readiness probe?
- 09 What is a rolling update?
- 10 What is Helm?
- 11 What is AWS EKS?
- 12 What is AWS RDS vs self-managed DB?
- 13 What is AWS ElastiCache and MSK?
- 14 What is IAM? What is a VPC?
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■ BEHAVIOURAL

- 01 Tell me about yourself
- 02 Walk me through your most complex project
- 03 Tell me about a production issue you solved
- 04 Tell me about a design decision you made and why
- 05 Tell me about a time you disagreed with a colleague
- 06 Tell me about a time you failed and what you learned
- 07 Where do you see yourself in 3 years?
- 08 Why are you leaving your current company?
- 09 Why do you want to join us?
- 10 What is your biggest technical weakness?
- 11 Tell me about a time you mentored someone
- 12 How do you handle tight deadlines?
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Interview Tips

Answer Formula

WHAT + HOW + WHY every time. Never stop at WHAT.

Practice Out Loud

Explain each concept daily without notes. Record yourself weekly.

Daily DSA

1 Leetcode medium every morning. Focus on patterns not memorization.

Feynman Technique

After each session close everything and explain what you built out loud.

Negotiate Always

Never accept first offer. Always have competing offers. 25 LPA → 30 LPA.

Your Advantage

You built this system from scratch. Most candidates just used someone else's code.

You built this. You understand it. Now go prove it. ■